

Project Testing

PocketSmart AI – Your Smart Budget & Recommendation Assistant

1. Testing Objective

Testing is performed to verify that PocketSmart AI works correctly across its different modules and produces relevant recommendations according to user requirements.

The major testing areas include:

- Functional testing
 - Input validation
 - AI response testing
 - Authentication testing
 - Session testing
 - UI testing
 - Budget adherence testing
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2. Functional Testing

Test Case 1 – User Registration

Input: Valid registration details

Expected Result:

A new user account should be created successfully.

Test Case 2 – User Login

Input: Valid username/email and password

Expected Result:

The user should be authenticated and redirected to the appropriate interface.

Test Case 3 – Invalid Login

Input: Incorrect credentials

Expected Result:

The system should reject the login attempt.

Test Case 4 – Home Planner**Input:**

- Budget
- Room type
- Product requirements

Expected Result:

The system should generate home interior recommendations.

Test Case 5 – Party Planner**Input:**

- Budget
- Guest count
- Event type

Expected Result:

The system should generate suitable food, venue and decoration recommendations.

Test Case 6 – Jewelry Planner**Input:**

- Budget
- Occasion
- Style
- Optional outfit image

Expected Result:

The system should generate jewelry suggestions based on the available information.

3. AI Testing

The AI recommendation engine is tested for:

Budget Adherence

Recommendations should remain relevant to the user's specified budget.

Context Awareness

The AI should consider the user's selected category and preferences.

Recommendation Quality

Generated products/services should be relevant to the user's requirements.

Image Understanding

The Jewelry Planner should be tested with suitable outfit images.

4. API Testing

The following endpoints should be tested:

```
/register  
/login  
/logout  
/token  
  
/generate-home  
/generate-party  
/generate-jewelry  
  
/session-info  
/session-data  
/recommendations-details  
/history
```

5. Edge Case Testing

The application should be tested with:

- Empty budget
- Very low budget
- Very high budget
- Missing required fields
- Invalid values

- Missing image
 - Unsupported image
 - Large guest count
 - Incomplete preferences
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6. Optimization

Testing results can be used to improve:

- AI prompts
- Input validation
- Error handling
- Session management
- UI/UX
- Recommendation quality

Fallback/default recommendations can be provided when the AI does not return sufficient results.

7. Expected Testing Result

The final system should:

- Accept valid inputs.
- Reject invalid inputs appropriately.
- Authenticate users.
- Process planner requests.
- Generate relevant AI recommendations.
- Maintain user sessions.
- Display results clearly.
- Preserve recommendation history.